S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

vallabh.pataneni@rwth-aachen.de
+49 17685944047
Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Master's student in Automotive Engineering with a **focused emphasis** on **design, simulation, and prototyping** of **sustainable automotive applications**. Possesses extensive experience in **generative design**, various **FEM software (PTC Creo to Ansys)**, **simulation tools (MATLAB/Simulink)**, and a **comprehensive understanding** of **different Motor technologies**, coupled with **advanced programming** and **proficient visualization skills**. Seeks a mandatory internship to further develop expertise in **hybrid transmission systems** and contribute to the central competences of the E-Mobility Division at Schaeffler.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems),Alternative Vehicle propulsion systems (ICE, BEV, HEV), FEM, DFM, Mobile Propulsion systems (E-motors, Fuel Cells, ICE).
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Suspension Division & member of Pitcrew 2025

- **Led the complete design cycle for suspension and steering systems**, from **conceptualization** and **optimization** through **Generative Design** and **extensive FEM** simulations leading to **production**. This involved implementing a **motorcycle-inspired trailing arm** and **double-wishbone front suspension** to optimize **driving dynamics** and minimize **rolling friction**.
- **Contributed to the structural engineering design and finite element analysis (FEA)** of a **hybrid Aluminum-carbon fiber chassis**, resulting in a **mass reduction** while maintaining structural integrity, demonstrating experience with meshed body analysis.
- **Developed and implemented data-driven driving strategies** using **Matlab-Simulink, and Power BI**, leveraging previous competition data to identify performance bottlenecks and **significantly improve overall vehicle efficiency**.
- **Contributed to the selection, integration design, and Assembly of the PMSM Wheel Hub Motor** for the car **Hydraix** in the Eco Shell Marathon competion 2025.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- led the end-to-end design and prototyping of a user-centric robot, focusing on **robot dynamics, Compactness** and **ease of repairability**.
- **Developed and implemented precise acceleration and deceleration curves** for robot motion by programming a **double H-bridge motor controller** using **Arduino IDE**, controlling a **NEMA 23 stepper motor** integrated with a **planetary gearbox**.
- Designed the **robot's enclosure** using forms design in **Fusion 360**, focusing on the neat integration of chassis components and overall **aesthetics**.
- Proficiently developed comprehensive **technical documentation**, managed diverse datasets, and delivered impactful presentations utilizing **MS Office**.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof of concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using a **Dual nozzle FDM based Rapid prototyping 3D printer (Raise 3D)**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks

Software Proficiency
Python
C/C++
MS Office Suite

Data Visualization Tools
Power BI
Tableau

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

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Summary

Highly motivated Master's student in Automotive Engineering with a strong focus on **computational mechanics**, **geometric modeling**, and **program development** for engineering applications. Possesses hands-on experience in **vehicle system design (Fusion 360)** and **data-driven strategy implementation (Power BI,Matlab/Simulink)**, complemented by proficiency in **simulation(FEM/FEA),prototyping and programming.** Seeking a mandatory internship to apply my analytical skills, technical understanding, and a systematic work style to support **product readiness for future requirements**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and Components,Material Science, Data Science, Fundamentals of Machine learning, Additive Manufaturing, FEM, DFM.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Wheel efficiency & member of Pitcrew 2025

- **Led the complete design cycle for suspension and steering systems**, from **conceptualization** and **optimization** through **Generative Design simulation (Fusion 360)** and **production**. This involved implementing a **motorcycle-inspired trailing arm** and **double-wishbone front suspension** to optimize **driving dynamics** and minimize **rolling friction**.
- **Contributed to the structural engineering design and finite element analysis (FEA)** of a **hybrid Aluminum-carbon fiber chassis**, resulting in a **mass reduction** while maintaining structural integrity, demonstrating experience with meshed body analysis.
- **Developed and implemented data-driven driving strategies** using **Matlab-Simulink, and Power BI**, leveraging previous competition data to identify performance bottlenecks and **significantly improve overall vehicle efficiency**.
- **Contributed to the design, integration, and maintenance of the gearbox for the motors of the "Hydraix" vehicle**, optimizing its torque transmission and overall drivetrain efficiency.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- led the end-to-end design and prototyping of a user-centric robot, focusing on **robot dynamics** and **iterative design and simulation**.
- **Developed and implemented precise acceleration and deceleration curves** for robot motion by programming a **double H-bridge motor controller** using **Arduino IDE**, controlling a **NEMA 23 stepper motor** integrated with a **planetary gearbox**.
- Successfully **implemented speech recognition AI functionality** for the doorbot, initially utilizing **Vosk** on **Raspberry Pi** and subsequently migrating to **Google API** for enhanced performance.
- **Leveraged Power BI to collect and analyze real-time data**, leading to **improvements in the efficiency of the doorbot's speech recognition** capabilities for command-based operation.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

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Skills

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Fusion 360
SolidWorks
PTC Creo

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks

Software Proficiency
Python
C/C++
MS Office Suite

Data Visualization Tools
Power BI
Tableau

Languages

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Academic Recognition
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Summary

Highly motivated Master's student in Automotive Engineering with a strong focus on **computational mechanics**, **geometric modeling**, and **program development** for engineering applications. Possesses hands-on experience in **vehicle system design (Fusion 360)** and **data-driven strategy implementation (Power BI,Matlab/Simulink)**, complemented by proficiency in **simulation(FEM/FEA),prototyping and programming.** Seeking a mandatory internship to apply my analytical skills, technical understanding, and a systematic work style to support **product readiness for future requirements**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and Components,Material Science, Data Science, Fundamentals of Machine learning, Additive Manufaturing, FEM, DFM.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
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Aachen, Germany

Ecogenium e.V
Head of Wheel efficiency & member of Pitcrew 2025

- **Led the complete design cycle for suspension and steering systems**, from **conceptualization** and **optimization** through **Generative Design simulation (Fusion 360)** and **production**. This involved implementing a **motorcycle-inspired trailing arm** and **double-wishbone front suspension** to optimize **driving dynamics** and minimize **rolling friction**.
- **Contributed to the structural engineering design and finite element analysis (FEA)** of a **hybrid Aluminum-carbon fiber chassis**, resulting in a **mass reduction** while maintaining structural integrity, demonstrating experience with meshed body analysis.
- **Developed and implemented data-driven driving strategies** using **Matlab-Simulink, and Power BI**, leveraging previous competition data to identify performance bottlenecks and **significantly improve overall vehicle efficiency**.
- **Contributed to the development and implementation of the vehicle's braking system**, encompassing **testing, evaluation, and routine maintenance activities** (e.g., bleeding brakes) to ensure optimal performance, functionality, and safety for competition.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- led the end-to-end design and prototyping of a user-centric robot, focusing on **robot dynamics** and **iterative design and simulation**.
- **Programmed a double H-bridge motor controller** using **Arduino IDE** to implement precise acceleration and deceleration curves for robot movement.
- Successfully **implemented speech recognition AI functionality** for the doorbot, initially utilizing **Vosk** on **Raspberry Pi** and subsequently migrating to **Google API** for enhanced performance.
- **Leveraged Power BI to collect and analyze real-time data**, leading to **improvements in the efficiency of the doorbot's speech recognition** capabilities for command-based operation.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof of concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using a **Dual nozzle FDM based Rapid prototyping 3D printer (Raise 3D)**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks

Software Proficiency
Python
C/C++
MS Office Suite

Data Visualization Tools
Power BI
Tableau

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

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linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering with a strong focus on **product design and performance evaluation, data analysis, and efficiency optimization**. Possesses hands-on experience in **vehicle system design (Fusion 360)** and **data-driven strategy implementation (Power BI,Matlab/Simulink)**, complemented by proficiency **simulation,prototyping and programming.** Seeking a mandatory internship to apply my analytical skills, technical understanding, and a systematic work style to support **product readiness for future requirements**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and Components,Material Science, Data Science, Fundamentals of Machine learning, Additive Manufaturing.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Wheel efficiency & member of Pitcrew 2025

- **Contributed to the development and implementation of the vehicle's braking system**, encompassing **testing, evaluation, and routine maintenance activities** (e.g., bleeding brakes) to ensure optimal performance, functionality, and safety for competition.
- **Led the complete design cycle for suspension and steering systems**, from **conceptualization** and **optimization** through **Generative Design simulation (Fusion 360)** and **production**. This involved implementing a **motorcycle-inspired trailing arm** and **double-wishbone front suspension** to optimize **driving dynamics** and minimize **rolling friction**.
- **Developed and implemented data-driven driving strategies** using **Python, Simulink, and Power BI**, leveraging previous competition data to identify performance bottlenecks and **significantly improve overall vehicle efficiency**.
- Proficiently developed comprehensive **technical documentation**, managed diverse **datasets**, and delivered impactful **presentations** utilizing MS Office.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- led the end-to-end design and prototyping of a user-centric robot, focusing on **robot dynamics** and **iterative design and simulation**.
- **Programmed a double H-bridge motor controller** using **Arduino IDE** to implement precise acceleration and deceleration curves for robot movement.
- Successfully **implemented speech recognition AI functionality** for the doorbot, initially utilizing **Vosk** on **Raspberry Pi** and subsequently migrating to **Google API** for enhanced performance.
- **Leveraged Power BI to collect and analyze real-time data**, leading to **improvements in the efficiency of the doorbot's speech recognition** capabilities for command-based operation.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof of concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using a **Dual nozzle FDM based Rapid prototyping 3D printer (Raise 3D)**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks

Software Proficiency
Python
C/C++
MS Office Suite

Data Visualization Tools
Power BI
Tableau

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

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Summary

Highly motivated Master's student in Automotive Engineering specializing in **vehicle dynamics** and **AI based testing methodologies**, with a strong interest in **wheel brake development**. Possesses practical experience in **braking system development, testing, evaluation, and maintenance**, coupled with proficient **programming skills (Python, MATLAB/Simulink)** and an eagerness to apply **AI algorithms in Automotive Technologies**. Seeks a Mandatory Internship to contribute to innovative brake concepts, performance testing, and data analysis within a dynamic team.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and Components, Additive Manufacturing, Fundamentals of Machine Learning.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Wheel efficiency & member of Pitcrew 2025

- **Contributed to the development and implementation of the vehicle's braking system**, encompassing **testing, evaluation, and routine maintenance activities** (e.g., bleeding brakes) to ensure optimal performance, functionality, and safety for competition.
- **Developed and implemented driving strategy algorithms** using **Python and Simulink** to enhance overall efficiency, demonstrating practical application of programming for system optimization and data-driven decisions.
- Led the complete design cycle for the **vehicle's suspension** and **steering systems**, optimizing **driving dynamics** and addressing complex challenges like turning radius and suspension stiffness.
- Contributed to the structural engineering design and **finite element analysis (FEA)** of a hybrid Aluminum-carbon fiber chassis, achieving a mass reduction while maintaining structural integrity.
- Managed d**river communication** and **race strategy** for the 2025 Eco-Marathon, coordinating the driver to achieve 5 valid runs and secure a **runners-up** position.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- ed the end-to-end design and prototyping of a user-centric robot, focusing on **robot dynamics** and **iterative design and simulation**.
- **Programmed a double H-bridge motor controller** using **Arduino IDE** to implement precise acceleration and deceleration curves for robot movement.
- Successfully **implemented speech recognition AI functionality** for the doorbot, initially utilizing **Vosk** on **Raspberry Pi** and subsequently migrating to **Google API** for enhanced performance.
- Operated FDM-based dual-nozzle 3D printers for rapid prototyping and fabrication of over 150 functional components.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using **FDM-based Rapid prototyping 3D printer**.

Skills

CAD Software
Catia V5
Fusion 360
SolidWorks
PTC Creo

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks

Rendering & Visualization
Keyshot
Rhino

Software Proficiency
Python
C/C++
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

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+49 17685944047
Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Master's student in Automotive Engineering, specialized in the **complete product development cycle from conceptualization and design through prototyping and manufacturing**. With a strong focus on **additive manufacturing**, particularly **metal 3D printing**, and **Composite Material technology**. Possesses practical experience in **experimentation, data analysis, and process development**, complemented by robust skills in **3D CAD design** and **production technology**. Seeking a Mandatory Internship at Robert Bosch GmbH to advance the development of metal 3D printing, contribute to experimental work, and collaborate on innovative process development within this cutting-edge field.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and design, Additive Manufacturing, DFM, Material Science, Alternative propulsions systems.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Suspension and Steering Division & member of Pitcrew 2025

- Led the **conceptualization and optimization** for the "Katara" vehicle's **suspension and steering systems**, addressing complex challenges in **turning radius** and **suspension stiffness** by implementing a **motorcycle-inspired trailing arm suspension** for the rear, optimizing **driving dynamics**.
- Executed the complete **design cycle** for these systems, utilizing **CATIA V5** for conceptualization and **Generative Design simulation (Fusion 360)** through to production **using metal 3D printing(DMLS).**
- Contributed to the design and **manufacturing** of "Katara's" **carbon fiber aeroshell** using the **prepreg layup method**, leveraging **flow simulation** for **aerodynamic performance refinement**.
- **Developed and implemented driving strategy algorithms** using **Python and Simulink** to enhance the **overall efficiency of the fuel cell system** during competition runs.
- **Managed driver communication and race strategy** for the 2025 Eco-Marathon, **coordinating the driver to achieve 5 valid runs** and secure a **runners-up position in the hydrogen fuel cell category**.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the **end-to-end design and prototyping** of a **user-centric robot (doorbot)**, focusing on **robot dynamics** and enhancing the overall **user experience** through **iterative design and simulation**.
- **Programmed a double H-bridge motor controller** using **Arduino IDE** to implement precise **acceleration and deceleration curves** for robot movement.
- Worked with **Raspberry Pi** to effectively control the **user interface** of the robot, ensuring seamless interaction.
- Operated **FDM-based dual-nozzle 3D printers (Raise 3D)** for **rapid prototyping and fabrication** of over 150 functional components across various engineering sub-projects.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using **FDM-based Rapid prototyping 3D printer**.

Skills

CAD Software
Catia V5
Fusion 360
SolidWorks
PTC Creo

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks

Rendering & Visualization
Keyshot
Rhino

Software Proficiency
Python
C/C++
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

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Summary

Master's student in Automotive Engineering, specialized in the **complete product development cycle from conceptualization and design through prototyping and manufacturing**. Possessing robust practical experience in **3D CAD design, advanced simulation, and cutting-edge production technologies**, complemented by foundational programming skills. Seeking a Mandatory Internship to apply comprehensive technical and software expertise to **mechanical design projects**, with a keen interest in contributing to the development and construction of research demonstrators and solving complex assembly and transport tasks at Robert Bosch GmbH, leveraging skills in **robotics, mechatronics, and 3D printing**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and design, FEM, DFM, Material Science, Robotics, Mechatronics.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Suspension and Steering Division & member of Pitcrew 2025

- Led the **conceptualization and optimization** for the "Katara" vehicle's **suspension and steering systems**, addressing complex challenges in **turning radius** and **suspension stiffness** by implementing a **motorcycle-inspired trailing arm suspension** for the rear, optimizing **driving dynamics**.
- Executed the complete **design cycle** for these systems, utilizing **CATIA V5** for conceptualization and **Generative Design simulation (Fusion 360)** through to production **using metal 3D printing.**
- Contributed to the design and **manufacturing** of "Katara's" **carbon fiber aeroshell** using the **prepreg layup method**, leveraging **flow simulation** for **aerodynamic performance refinement**.
- Played a key role in the **manufacturing** of the **carbon fiber steering wheel** using the **resin infusion method** and engineered a **fuel cell system rig** with **carbon fiber components**, optimizing **packaging** and integration.
- **Developed and implemented driving strategy algorithms** using **Python and Simulink** to enhance the **overall efficiency of the fuel cell system** during competition runs.
- **Managed driver communication and race strategy** for the 2025 Eco-Marathon, **coordinating the driver to achieve 5 valid runs** and secure a **runners-up position in the hydrogen fuel cell category**.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the **end-to-end design and prototyping** of a **user-centric robot (doorbot)**, focusing on **robot dynamics** and enhancing the overall **user experience** through **iterative design and simulation**.
- **Programmed a double H-bridge motor controller** using **Arduino IDE** to implement precise **acceleration and deceleration curves** for robot movement.
- Worked with **Raspberry Pi** to effectively control the **user interface** of the robot, ensuring seamless interaction.
- Operated **FDM-based dual-nozzle 3D printers (Raise 3D)** for **rapid prototyping and fabrication** of over 150 functional components across various engineering sub-projects.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity**.

Skills

CAD Software
Catia V5
Fusion 360
SolidWorks
PTC Creo
Siemens NX

Simulation Software
Ansys
ANSA
Matlab - Simulink
Fusion 360
Solidworks
CADFEM

Rendering & Visualization
Keyshot
Rhino
Adobe illustrator

Software Proficiency
Python
C/C++
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

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International Journal of Science & Engineering Development Research
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linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering with expertise spanning the **entire vehicle development lifecycle, from initial conceptualization and design through prototyping and manufacturing**. Specializing in **chassis, aeroshell and suspension design** along with various cutting edge **manufacturing techniques (Resin infusion CFRP method, Metal 3D printing)** for Urban concept vehicles. Proficient in **mechanical and generative design, dynamic simulation, and technical drawing** using advanced **CAD systems (CATIA V5, Fusion 360)**. Seeking a mandatory internship to leverage my strong **technical understanding of motorcycles** and a collaborative spirit for innovative design projects.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Dynamics and design, FEM, DFM, Material Science, Flevo Bike Design and development.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Suspension and Steering Division & member of Pitcrew 2025

- Led the **conceptualization and optimization** for the "Katara" vehicle's **suspension and steering systems**, addressing complex challenges in **turning radius** and **suspension stiffness** by implementing a **motorcycle-inspired trailing arm suspension** for the rear, optimizing **driving dynamics**.
- Executed the complete **design cycle** for these systems, utilizing **CATIA V5** for conceptualization and **Generative Design simulation (Fusion 360)** through to production **using metal 3D printing.**
- Participated in the **structural engineering design and finite element analysis (FEA)** of a **hybrid Aluminum-carbon fiber chassis**, resulting in a **mass reduction of 25 kg** while maintaining **structural integrity**.
- Contributed to the design and **manufacturing** of "Katara's" **carbon fiber aeroshell** using the **prepreg layup method**, leveraging **flow simulation** for **aerodynamic performance refinement**.
- Played a key role in the **manufacturing** of the **carbon fiber steering wheel** using the **resin infusion method** and engineered a **fuel cell system rig** with **carbon fiber components**, optimizing **packaging** and integration.
- **Managed driver communication and race strategy** for the 2025 Eco-Marathon, **coordinating the driver to achieve 5 valid runs** and secure a **runners-up position in the hydrogen fuel cell category**.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the **end-to-end design and prototyping** of a **user-centric robot (doorbot)**, focusing on **robot dynamics** and enhancing the overall **user experience** through **iterative design and simulation**.
- Executed the **industrial design** for the robot's **exterior shell**, prioritizing **compactness** and **refined aesthetics** while skillfully **integrating complex internal components (sensors, PCBs)** and ensuring **intuitive interface design** within **spatial constraints**.
- Operated **FDM-based dual-nozzle 3D printers (Raise 3D)** for **rapid prototyping and fabrication** of over 150 functional components across various engineering sub-projects.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity**.

Skills

CAD Software
Catia V5
Fusion 360
SolidWorks
PTC Creo
Siemens NX

Simulation Software
Ansys
ANSA
Matlab - Simulink
Fusion 360
Solidworks
CADFEM

Rendering & Visualization
Keyshot
Rhino
Adobe illustrator

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

vallabh.pataneni@rwth-aachen.de

+49 17685944047

Bahnhofstraße 59/2, Renningen, Germany

linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering with a strong focus on **fiber-reinforced composite materials, precision mechanical design,** and **advanced manufacturing processes**. Proven hands-on experience in **carbon fiber fabrication (e.g., resin infusion, prepreg layup)**, CAD-based engineering (**PTC Creo, Siemens NX, Fusion 360**), and experimental and simulation based analysis(**Ansys**, **CADFEM**).Seeking a mandatory internship to leverage my skills in **Production engineering, reverse engineering, and composite component overhaul** for high-performance aviation and Automobile industries.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Optimization of light-weight systems,Material science, 3D modelling, FEM, DFM.Automotive Dynamics and design.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Head of Suspension and Steering Division & member of Pitcrew 2025

- **Led the complete design cycle for the "Katara" Eco-Marathon vehicle's suspension systems**, from conceptualization and optimization through Generative Design and structural simulation (**PTC Creo, ANSYS**) and production using **metal 3D printing**.
- **Contributed significantly to the holistic design and manufacturing of "Katara's" carbon fiber aeroshell.** Focused on the critical interplay between **aerodynamics and exterior aesthetics**, leveraging **flow simulation** for performance refinement. **Actively participated in the manufacturing of the aeroshell using the prepreg layup method**, gaining direct experience with advanced composite fabrication.
- Played a key role in the **manufacturing of the carbon fiber steering wheel using the resin infusion method**.
- As part of the **pit crew**, regularly performed **Non-Destructive Testing (NDT)** to identify structural defects in composite components (mainly chassis) and conducted routine **maintenance activities** (e.g., greasing, bleeding brakes) to ensure vehicle integrity and performance.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the **design and manufacturing of a user-centric robot (doorbot)**, focusing on **Compact packaging** and developing its **visually appealing exterior shell** through **iterative design and prototyping**.
- Conducted **routine maintenance** on the **robot's gearboxes** and other moving components to ensure **optimal performance** and **longevity**.
- **Utilized FDM-based dual-nozzle 3D printers** (Raise 3D) for **rapid prototyping and fabrication of various composites with continuous fibres (Reinforced glass fibre composites)**, critically **supporting the visualization and increasing the strength of the components.**

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX
Catia V5

Rendering & Visualization
Keyshot
Rhino
Adobe illustrator

Simulation Software
Ansys
ANSA
Matlab - Simulink
Fusion 360
Solidworks
CADFEM

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

vallabh.pataneni@rwth-aachen.de
+49 17685944047
Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Master's student in Automotive Engineering (RWTH Aachen University) bringing a unique blend of **vehicle development expertise** and a strong focus on **3D visualization and design communication**. With proven experience in **generative design, rapid prototyping, and design optimization** (including structural/thermal simulation and topology optimization), I possess a clear **sense of aesthetics**. Seeking a mandatory internship to leverage these skills in creating **appealing and contemporary visualizations** for **vehicle body design and development**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Industrial design Engineering, Ergonomics, 3D modelling and animation.Automotive Dynamics and design.

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division

- **Led the complete design cycle** for the "Katara" Eco-Marathon vehicle's suspension systems, from **conceptualization** and optimization through **Generative Design simulation and production**. Crucially, utilized **Fusion 360 and Keyshot for high-quality visualization** of the complex structures, demonstrating a methodical approach to engineering and design.
- **Contributed to the holistic design of "Katara's" aeroshell**, focusing on the critical **interplay between aerodynamics and exterior aesthetics**, and leveraging **flow simulation to refine both performance and visual appeal**.
- **Gained comprehensive insight into the entire vehicle development process**, encompassing **interior/exterior aesthetics, driver ergonomics, and intuitive vehicle communication**, providing a strong foundation for **visualizing and animating complex vehicle concepts** from idea to realization.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the **conceptualization and design of a user-centric robot (doorbot)**, focusing on **enhancing the overall user experience (UX)** and developing its **visually appealing exterior shell** through **iterative design and prototyping**.
- **Executed the industrial design for the robot's exterior shell**, prioritizing **refined aesthetics and compactness** while **skillfully integrating complex internal components** and ensuring an **intuitive interface design** within **spatial constraints**.
- **Utilized FDM-based dual-nozzle 3D printers** (Raise 3D) for **rapid prototyping and fabricating over 150 functional components**, critically **supporting the visualization and iterative refinement of designs** across various projects.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX
Catia V5

Simulation Software
Ansys
ANSA
Matlab - Simulink
Fusion 360
Solidworks

Rendering & Visualization
Keyshot
Rhino
Adobe illustrator

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

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Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering (RWTH Aachen University) with proven experience in **generative design** and **vehicle development**. Passionate about **rapid prototyping** and **design optimization**, I offer **strong technical skills** in **Vehicle design**, **structural** and **thermal simulation, topology optimization** and **additive manufacturing**. Seeking a **mandatory internship** to apply a user-centered design approach and enhance expertise in **Vehicle Design** and **Development**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Vehicle Design, Vehicle Dynamics, FEM,Additve Manufacturing,Material Science,GT&T, DFM.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division

- Led the full cycle of **structural design, multi-objective optimization, and FEA simulation** using **Fusion 360** and **ANSYS** for the **double wishbone and trailing-arm suspension systems** of a hydrogen powered Eco-Marathon vehicle.
- Oversaw the **carbon fiber steering wheel fabrication (optimized for stiffness/weight)** using Resin Infusion method and also the **production** of **Front** and **rear suspension parts** via **Metal 3D printing** using **DMLS** method.
- Contributed to **aerodynamic design and CFD simulation** for vehicle's aeroshell, focusing on optimal component packing and **maneuverability optimization** using fusion 360.
- **Developed and simulated complex driving strategies in MATLAB and Simulink** to **optimize vehicle operational efficiency** and enhance **fuel cell performance** through start-up operation improvements.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the end-to-end design and **prototyping** of a user-centric robot (doorbot), focusing on **robot dynamics** and enhancing the overall user experience through **iterative design** and **simulation.**
- Executed the industrial design for the robot's exterior shell, prioritizing compactness and refined aesthetics while skillfully **integrating complex internal components (sensors, PCBs)** and ensuring **intuitive interface design** within **spatial constraints**.
- **Operated FDM-based dual-nozzle 3D printers** (Raise 3D) for **rapid prototyping** and **fabrication** of over **150 functional components** across various **engineering sub-projects**.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX
Catia V5

Simulation Software
Ansys
ANSA
Matlab - Simulink
Fusion 360
Solidworks

Rendering & Visualization
Keyshot
Rhino
Adobe illustrator

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

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Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering (**RWTH Aachen University**) with **proven experience** in **industrial design** and vehicle development. Passionate about "Generative Design", I offer **strong technical skills**, **expertise** in **Rhino, SolidWorks, Keyshot** and various **design** and **simulation** softwares. I'm seeking a **mandatory six-month internship** to contribute to diverse projects to **test my industrial design skills**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Alternative Vehicle propulsion systems, Fuel cell system technology, Vehicle testing and analysis, Material Science, FEM.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division

- **Led the complete design cycle** for the "Katara" Eco-Marathon vehicle's **suspension systems**, from **conceptualization and optimization** (using SolidWorks and Keyshot for visualisation) through **Generative design simulation** and **metal 3D printing for production**, demonstrating a methodical approach to complex engineering challenges.
- **Contributed to the holistic design of "Katara's" aeroshell**, focusing on the critical interplay between **aerodynamics and exterior aesthetics**, and leveraging **flow simulation** to refine performance and visual appeal.
- **Gained comprehensive insight into vehicle development,** encompassing **interior** and **exterior aesthetics, driver ergonomics, and intuitive vehicle communication,** to ensure a cohesive and user-centric overall experience.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the end-to-end design and **prototyping** of a user-centric robot (doorbot), focusing on **robot dynamics** and enhancing the overall user experience through **iterative design** and **simulation in Fusion 360**.
- Executed the industrial design for the robot's exterior shell, prioritizing compactness and refined aesthetics while skillfully **integrating complex internal components (sensors, PCBs)** and ensuring **intuitive interface design** within **spatial constraints**.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof-of-concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- **Executed** the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity**.

Skills

CAD Software
SolidWorks
PTC Creo
Fusion 360
Siemens NX
Catia V5

Simulation Software
Solidworks
PTC Creo
Ansys
Fusion 360
Matlab - Simulink

Rendering & Visualization
Keyshot
Rhino
Adobe illustrator

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

vallabh.pataneni@rwth-aachen.de
+49 17685944047
Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering adept at the full engineering lifecycle, from **conceptual design and advanced simulation to rigorous experimental testing and rapid prototyping**. Specializing in **fuel cell systems, thermodynamic analysis, and data evaluation**, with proven experience in hydrogen fuel cell compatibility and experimental work.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Alternative Vehicle propulsion systems, Fuel cell system technology, Vehicle testing and analysis, Material Science, FEM.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division & member of vehicle operations team

- As part of the Vehicle Operations team, I performed experimental analysis (e.g, Humidity cycling, purging) and data evaluation to optimize fuel cell performance and integration within the vehicle.
- Designed and fabricated a compact fuel cell test rig and a water-cooling system, optimizing component packaging density for fuel cells and associated electronic systems through detailed design within PTC Creo.
- Led the design, optimization, simulation of front (double-wishbone) and rear (trailing-arm) suspension systems for a hydrogen-powered Eco-Marathon vehicle using PTC Creo.
- Oversaw carbon fiber steering wheel manufacturing using Resin infusion method, optimizing layup for stiffness/weight.
- Oversaw the manufacturing of Front and rear suspension parts (Shape optimized) using Metal 3D printers.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Designed, simulated, and prototyped a door-opening robot, focusing on robot dynamics and user experience using Fusion 360.
- Integrated sensors and PCBs, considering spatial constraints and interface design.
- Executed thermal simulations for diverse power-electronic components, designing custom thermal management solutions (heatsinks) for PCBs.
- Developed, tested, and integrated a robust noise cancellation system utilizing a dual-microphone array.

Skills

CAD Software
PTC Creo
Fusion 360
SolidWorks
Siemens NX
Catia V5

Rendering & Visualization
Blender
Keyshot

Simulation Software
PTC Creo
Ansys
Solidworks
Fusion 360
Matlab - Simulink

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

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linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in **Automotive Engineering** at **RWTH Aachen University** with expertise in designing and prototyping vehicle systems for **Green Mobility**. Proven skills in **Vehicle operations** , **Simulation** and **Experimental analysis**. Proficient in **data evaluation, system analysis (MATLAB/Simulink),** and problem-solving for sustainable transportation solutions.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Vehicle Design, Alternative Vehicle propulsion systems, Vehicle testing and analysis, Material Science, FEM.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division & member of vehicle operations team

- As part of the Vehicle Operations team, I performed experimental analysis (e.g, Humidity cycling, purging) and data evaluation to optimize fuel cell performance and integration within the vehicle, identifying and addressing technical challenges.
- Designed and fabricated a compact fuel cell test rig, optimizing component packaging density for fuel cells and associated electronic systems through detailed design within PTC Creo.
- Oversaw carbon fiber steering wheel manufacturing using Resin infusion method, optimizing layup for stiffness/weight.
- Oversaw the manufacturing of Front and rear suspension parts (Shape optimized) using Metal 3D printers.
- Led the design, optimization, simulation of front (double-wishbone) and rear (trailing-arm) suspension systems for a hydrogen-powered Eco-Marathon vehicle using PTC Creo.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Designed, simulated, and prototyped a door-opening robot, focusing on robot dynamics and user experience using Fusion 360.
- Integrated sensors and PCBs, considering spatial constraints and interface design.
- Selected motors and implemented motor cooling strategies.
- Developed, tested, and integrated a robust noise cancellation system utilizing a dual-microphone array.
- Designed and fabricated shell enclosure, focusing on aesthetics.

Skills

CAD Software
PTC Creo
Fusion 360
SolidWorks
Siemens NX
Catia V5

Rendering & Visualization
Blender
Keyshot

Simulation Software
PTC Creo
Ansys
Solidworks
Fusion 360
Matlab - Simulink

Software Proficiency
Python
MS Office Suite

Languages

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Bahnhofstraße 59/2, Renningen, Germany

linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in **Automotive Engineering** at **RWTH Aachen University** with expertise in designing and prototyping vehicle systems for **Green Mobility**. Proficiency in comprehensive **CAD development**, **Simulation** and **Rapid Prototyping**. My proactive approach to design and robust simulation capabilities enable me to translate complex concepts into **functional prototypes**, driving innovation in vehicle systems.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Vehicle Design, Automated and connected Driving challanges, Vehicle Dynamics, Material Science, DFM, FEM.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division

- Led the design, optimization, simulation of front (double-wishbone) and rear (trailing-arm) suspension systems for a hydrogen-powered Eco-Marathon vehicle using PTC Creo.
- Designed and fabricated a compact fuel cell test rig, optimizing component packaging density for fuel cells and associated electronic systems through detailed design within PTC Creo.
- Oversaw carbon fiber steering wheel manufacturing using Resin infusion method, optimizing layup for stiffness/weight.
- Oversaw the manufacturing of Front and rear suspension parts (Shape optimized) using Metal 3D printers.
- Delivered technical presentations and generated supporting documentation using MS office.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Designed, simulated, and prototyped a door-opening robot, focusing on robot dynamics and user experience using Fusion 360.
- Integrated sensors and PCBs, considering spatial constraints and interface design.
- Selected motors and implemented motor cooling strategies.
- Developed robot chassis, emphasizing smart packaging and component protection.
- Designed and fabricated shell enclosure, focusing on aesthetics.

01/2023 – 03/2023
Hyderabad

Industrial Design Intern
Garuda 3D

- Contributed to the design and prototyping of a scaled-down air taxi for proof of concept using Solidworks.
- Design, simulation and fabrication of Air taxi wings focusing on maneourablity.
- Design and simulation of Landing mechanism.

Skills

CAD Software
PTC Creo
Fusion 360
SolidWorks
Siemens NX
Catia V5

Simulation Software
PTC Creo
Ansys
Solidworks
Fusion 360
Matlab - Simulink

Rendering & Visualization
Blender
Keyshot

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

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Master's student in Automotive Engineering

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Bahnhofstraße 59/2, Renningen, Germany | linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering at RWTH Aachen University with a strong focus on **Green-mobility** and extensive experience in vehicle development projects. Proven skills in **CAD**, **Simulation**, and **Prototyping** with foundational programming skills, seeking to apply technical expertise and contribute to **Mechanical design projects** within a dynamic internship. Eager to engage in the full spectrum of **Product development**, from concept to implementation, and collaborate within an interdisciplinary team.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Vehicle Design, Optimization of light weight designs, Finite element methods, Fundamentals of maching learning.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division

- Led the design, optimization, simulation of front (double-wishbone) and rear (trailing-arm) suspension systems for a hydrogen-powered Eco-Marathon vehicle using PTC Creo.
- Oversaw carbon fiber steering wheel manufacturing using Resin infusion method, optimizing layup for stiffness/weight.
- Oversaw the manufacturing of Front and rear suspension parts (Shape optimized) using Metal 3D printers.
- Contributed to aeroshell design and flow simulation.
- Delivered technical presentations and generated supporting documentation using MS office.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Designed, simulated, and prototyped a door-opening robot, focusing on robot dynamics and user experience using Fusion 360.
- Integrated sensors and PCBs, considering spatial constraints and interface design.
- Selected motors and implemented motor cooling strategies.
- Developed robot chassis, emphasizing smart packaging and component protection.
- Designed and fabricated shell enclosure, focusing on aesthetics.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- Contributed to the design and prototyping of a scaled-down air taxi for proof of concept using Solidworks.
- Design, simulation and fabrication of Air taxi wings focusing on maneourablity.
- Design and simulation of Landing mechanism.

Skills

CAD Software
PTC Creo
Fusion 360
SolidWorks
Siemens NX
Catia V5

Simulation Software
PTC Creo
Ansys
Solidworks
Fusion 360
Matlab - Simulink

Rendering & Visualization
Blender
Keyshot

Software Proficiency
Python
MS Office Suite

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

vallabh.pataneni@rwth-aachen.de
+49 17685944047
Bahnhofstraße 59/2, Renningen, Germany
linkedin.com/in/vallabh-pataneni

Summary

Highly motivated Master's student in Automotive Engineering at RWTH Aachen University with a strong focus on **Green-mobility** and extensive experience in vehicle development projects. Proven skills in **CAD**, **Simulation**, and **Prototyping**, seeking to apply technical expertise and contribute to **Mechanical design projects** within a dynamic internship. Eager to engage in the full spectrum of **Product development**, from concept to implementation, and collaborate within an interdisciplinary team.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Vehicle Design, Optimization of light weight designs, Alternative Vehicle Propulsion systems, GD&T, Vehicle Dynamics, DFM.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen

Ecogenium e.V
Head of Suspension and Steering Division

- Led the design, optimization, simulation of front (double-wishbone) and rear (trailing-arm) suspension systems for a hydrogen-powered Eco-Marathon vehicle using PTC Creo.
- Oversaw carbon fiber steering wheel manufacturing using Resin infusion method, optimizing layup for stiffness/weight.
- Oversaw the manufacturing of Front and rear suspension parts (Shape optimized) using Metal 3D printers.
- Contributed to aeroshell design and flow simulation.
- Delivered technical presentations and generated supporting documentation using MS office.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Designed, simulated, and prototyped a door-opening robot, focusing on robot dynamics and user experience using Fusion 360.
- Integrated sensors and PCBs, considering spatial constraints and interface design.
- Selected motors and implemented motor cooling strategies.
- Developed robot chassis, emphasizing smart packaging and component protection.
- Designed and fabricated shell enclosure, focusing on aesthetics.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

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Highly motivated Master's student in Automotive Engineering at RWTH Aachen University with a strong focus on **Green-mobility** and extensive experience in vehicle development projects. Proven skills in **design**, **simulation**, and **prototyping**, seeking to apply technical expertise and contribute to innovative E-vehicle development within a dynamic internship. Eager to engage in the full spectrum of **vehicle development**, from concept to implementation, and collaborate within an interdisciplinary team.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Vehicle Design, Battery storage systems, Alternative Vehicle Propulsion systems, Intellectual property policy and law,Vehicle Dynamics.
GPA: 1.7 - (Currently in 4th Semester)

06/2019 – 04/2023
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Summary

Highly motivated Master's student in Automotive Engineering at RWTH Aachen University with a strong passion for automotive design and proven experience in vehicle development and industrial design projects. Seeking to leverage technical and design skills in a dynamic Industrial Design Internship to contribute to innovative vehicle design and enhance user experience. Proficient in numerous CAD and simulation softwares, with experience in design optimization, prototyping, and manufacturing. Eager to apply creativity and a user-centered design approach to contribute to the world’s automotive industry.

Education

03/2026
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vallabh.pataneni@rwth-aachen.de +49 17685944047 Bahnhofstraße 59/2, Renningen

linkedin.com/in/vallabh-pataneni

Summary

Multidisciplinary automotive engineering Master's student seeking a mandatory internship at **Daimler Truck GMBH**, driven by the ambition to build a fully **autonomous car**. Highly experienced in **Python**, skilled with a wide range of **Python libraries**, **Power BI**, and **Tableau**. Adept at advanced **data science**, **system design**, and **leadership** in innovative vehicle technology projects. Proficient at developing robust **data pipelines**, applying **machine learning** and **computer vision**, and delivering **actionable insights** for both engineering and business stakeholders. Committed to advancing intelligent, autonomous vehicles through **teamwork**, continuous improvement, and a passion for turning **complex sensor data** into practical mobility solutions.

Education

Master of Science in Automotive Engineering 03/2026 | Aachen, Germany

RWTH Aachen University

Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems),Data Science, Fundamentals of Machine Learning, Deep Learning.

GPA : 1.7

Bachelors in technology - Mechanical Engineering 06/2019 – 04/2023 | Hyderabad, India

Jawaharlal Nehru Technological Univerisity

GPA: 1.3

Experience

Industrial Design Intern 03/2023 – 07/2023 | Hyderabad

Techolution

- Designed **end-to-end data logging pipelines** for audio and image data, enabling robust **time-stamped acquisition** for the **Robot DoorBot 3.0**.
- Developed an **automated speech recognition AI** using the **Google Speech-to-Text API**, with advanced **audio preprocessing**, **feature extraction**, and **voice activity detection** in **Python** to reliably trigger door automation on voice command.
- Set up and calibrated a **robotics test rig**, establishing **synchronized camera data acquisition** and **real-time event logging** to train and validate object manipulation tasks.
- Created and maintained a **data dashboard** for **camera image curation**, annotation, and quality monitoring, supporting **supervised learning** and **data pipeline automation**.
- Trained and evaluated **CNN models** for robotic screw-picking, developing **image augmentation strategies** and **performance metrics** for deployment-ready object detection. Employed multiple **Python libraries** throughout the project lifecycle.

Academic Project

Vehicle Sensor Data analysis Project - IKA 04/2025 – 07/2025 | Aachen

Autonomous Engineer

- Designed and built **Python data pipelines** to process **camera, LiDAR, and radar sensor data**, using **signal processing** and **feature extraction** for **environment mapping** and **behavior analysis**.
- Developed and deployed a **real-time traffic sign recognition system** utilizing **convolutional neural networks** and **transfer learning** on **GTSRB** and custom datasets for accurate **multi-class detection**.
- Implemented **connected driving features** by syncing **real-time traffic light status** with **vehicle telemetry** and developing **MPC-based trajectory optimization algorithms** for **intersection management**.
- Created interactive **Power BI dashboards** to **visualize** and **analyze large-scale vehicle time-series data**, supporting **anomaly detection** and **actionable insights**.

Student Club

Ecogenium e.V 06/2024 – Present | Aachen, Germany

Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the **Shell Eco-Marathon**, managing the complete **lifecycle from concept to competition ready vehicle.**
- Utilized **advanced data analysis** and **Power BI** for visualization of complex powertrain data, transforming raw vehicle data into **actionable insights** for technical and business stakeholders, with extensive Python-based processing.
- Developed and iterated **optimized driving strategies** and control algorithms, validated with simulations, which enabled data-driven decisions and transparent communication, resulting in an **80 percent increase in efficiency** and a **second-place finish in a national competition**.
- Proficient in **design, simulation, and prototyping**: created **CAD models**, **digital twins**, and high-fidelity simulations for design validation and prototyping.

Skills

CAD Software
Catia V5
3DExperience Solidworks
Fusion 360
PTC Creo
Siemens NX
Pro Engineering

Data Visualization Tools
Power BI
Tableau

Cloud
Azure
Data Bricks

Software Proficiency
Python
C/C++
MS Office Suite

Python Libraries
NumPy
Pandas
scikit-learn
PyTorch
TensorFlow

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

Academic Recognition
RWTH

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Paper Publication
International Journal of Science & Engineering Development Research

A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

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Master's student in Automotive Engineering -RWTH Aachen

vallabh.pataneni@rwth-aachen.de
+49 17685944047
Bahnhofstraße 59/2, Renningen
linkedin.com/in/vallabh-pataneni

Summary

A highly driven **automotive engineer** from **RWTH Aachen University** seeking a **mandatory internship** at **Tesla**, passionate about shaping **next-generation mobility**. Brings a strong technical foundation in **CAD modeling** with proficiency in **CATIA V5**, **Fusion 360**, and **Solidworks 3D Experience**, as well as extensive **simulation expertise** in tools such as **ANSYS**, **Solidworks CFD** and **MATLAB/Simulink**. In addition, offers skills in **generative design** and **advanced prototyping techniques** including **DMLS metal 3D printing** and **carbon fiber layup**. Renowned for transforming **creative concepts** into **manufacturable, high-performance solutions** through innovative and practical design. Demonstrates proven **leadership** by successfully coordinating **multidisciplinary projects**, excelling in clear **communication** across engineering and manufacturing teams, and consistently delivering **innovative results** within strict deadlines.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems),DFM, FEA, Material Science, Additive Manufacturing.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the **Shell Eco-Marathon**, managing the complete **lifecycle from concept to competition ready vehicle.**
- **Performed** the complete **development cycle** for the vehicle's **suspension systems**—engineering a **double wishbone suspension** at the front and a **trailing arm concept** at the rear. This included **iterative and generative CAD development (Catia V5)** , **kinematics and structural analysis**, and **prototyping** with **DMLS Printing** ensuring **optimal handling, weight distribution, and structural integrity** for competition performance.
- Collaborated on the modeling and production of an **ergonomic, lightweight steering wheel** using **resin-infused carbon fiber layup**, balancing **performance** and **tactile driver experience**.
- Contributed to **hybrid aluminum-carbon chassis development**, utilizing **prepreg carbon fiber sandwich layup with custom metal molds** to **maximize stiffness** and **occupant safety** while achieving **minimal weight**.
- **Designed and performed CFD simulations** of the **carbon fiber aeroshell** in **3DExperience Flow Simulation**, leading to a **12% reduction in drag coefficient** through **iterative, data-driven refinement**.
- Produced **technical documentation**, led **milestone reviews**, and presented **innovative, manufacturable designs** to **sponsors and partners**.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- Led the end-to-end design and mechanical development of a **user-centric door-opening robot**, applying **Fusion 360** for advanced CAD modeling of **complex parts and assemblies** under tight packaging constraints.
- Engineered **enclosures and component layouts** balancing **aesthetics**, **protection**, and **ergonomic user interaction.**
- Programmed **double H-bridge motor controllers (Arduino IDE)** for **precise motion control** and integrated **sensor arrays** to ensure robust and reliable performance in electromechanical systems.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- Contributed to the conceptualization and prototyping of a **proof-of-concept air taxi** (scaled model) using **SolidWorks**, with a focus on translating advanced mobility ideas into functional designs.
- Developed precise **CAD models** for complex flight-critical components such as **wings** and the **landing mechanism**, balancing **structural integrity** with **lightweight design** through iterative simulation.
- Performed **CFD simulations** on **fuselage** and **multiple aerofoil designs** to optimize geometry for improved **lift** and minimized **drag**.
- Utilized a **dual-nozzle FDM rapid prototyping 3D printer (Raise 3D)** for fabrication and continuous design refinement based on testing feedback.

Skills

CAD Software
Catia V5
3DExperience Solidworks
Fusion 360
PTC Creo
Siemens NX
Pro Engineering

Software Proficiency
Python
C/C++
MS Office Suite

Simulation Software
Ansys Structural & Fluent
Solidworks
Converge
Matlab - Simulink
Fusion 360

Data Visualization Tools
Power BI
Tableau

Languages

English
Fluent (C1)

German
Intermediate (B1)

Recognitions and Publications

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Malmedyer str 20, Aachen
linkedin.com/in/vallabh-pataneni

Summary

Enthusiastic **automotive engineer** with a **passion** for turning complex **vehicle technology** into seamless, real world **performance**. Currently seeking a hands-on **Mandatory internship** with **DEKRA Automobil GmbH** to apply my blend of **technical expertise**, **data driven design**, and practical **teamwork**. I thrive in **workshop environments** where **knowledge meets application** and every day brings a new **mechanical challenge**. My journey has ranged from **prototyping racecars** to **optimizing drone aerodynamics**, always with a focus on **learning**, **communication**, and **innovation**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems),Alternative Vehicle Propulsion systems,Internal Combustion Engines, Additive manufacturing.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen fuel cell powered vehicle** from scratch for the **Shell Eco-Marathon**, managing the complete **lifecycle from concept to competition ready vehicle.**
- **Led the simulation-driven design process** for **Katara's aeroshell**, **iteratively testing and refining** surface contours and component alignment to **optimize aerodynamic performance**, which resulted in a **12% reduction in overall drag coefficient**.
- **Analyzed wake dynamics, flow separation, and pressure distribution**, delivering **actionable recommendations** that **enhanced airflow efficiency** and led to measurable improvements in **competition rollout energy efficiency**.
- **Performed the full design, simulation, prototyping, manufacturing, and validation cycle** for the vehicle's **suspension systems**—engineering a **double wishbone suspension** at the front and a **trailing arm concept** at the rear. This included **iterative CAD development**, **kinematics and structural analysis**, and **physical fabrication** ensuring **optimal handling, weight distribution, and structural integrity** for competition performance.
- **Authored technical documentation**, **delivered technical presentations**; coordinated with **multiple teams** and **external partners** for **project synergy**.
- **Directed on-track troubleshooting**, **damage checks**, and **rapid repairs** under competition deadlines.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- **Led the end-to-end design and prototyping** of a **user-centric robot**, focusing on **robot dynamics** and **iterative design/simulation**.
- **Designed the robot's enclosure** ensuring **visual aesthetics**, **robust protection**, **optimal component integration**, and **ease of assembly**.
- **Designed and developed** a dedicated **test bench setup** to capture **visual data** for the purpose of training an **AI model** to guide a **robotic arm** in **screw-picking tasks**.

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- Designed and tested **scale air-taxi prototypes**, exploring **lift**, **stability**, and **structural integration**.
- **Executed CFD simulations** on **fuselage** and multiple **aerofoil designs**, shaping **geometry** for **drag/lift optimization**.
- Sourced **rapid fabrication** and performed **qualitative/quantitative validation** on **wind-tunnel models**.
- Performed the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using a **Dual nozzle FDM based Rapid prototyping 3D printer (Raise 3D)**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX
Pro Engineering

Simulation Software
Ansys Fluent
Converge
Openfoam (Basic)
Matlab - Simulink
Fusion 360
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Summary

Purposeful **automotive engineer** actively **seeking a mandatory internship** to further my journey into **motorsport** and advanced **powertrain engineering** with **TOYOTA GAZOO Racing Europe**. With direct experience spanning the **complete cycle** from **design** and **simulation** (**200+ hours** using leading **CFD platforms**) to hands-on **prototyping** and **manufacturing**—I thrive in environments where ideas quickly become working technology. My background combines meticulous **technical execution** with proven **team management**, leading multidisciplinary groups through complex vehicle development projects. Guided by a strong **passion for racing** and progress, I am eager to contribute my **engineering** and **CFD skills** to **TOYOTA GAZOO's high performance team**.

Education

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- **Led the simulation-driven design process** for **Katara's aeroshell**, **iteratively testing and refining** surface contours and component alignment to **optimize aerodynamic performance**, which resulted in a **12% reduction in overall drag coefficient**.
- **Analyzed wake dynamics, flow separation, and pressure distribution**, delivering **actionable recommendations** that **enhanced airflow efficiency** and led to measurable improvements in **competition rollout energy efficiency**.
- **Performed the full design, simulation, prototyping, manufacturing, and validation cycle** for the vehicle's **suspension systems**—engineering a **double wishbone suspension** at the front and a **trailing arm concept** at the rear. This included **iterative CAD development**, **kinematics and structural analysis**, and **physical fabrication** ensuring **optimal handling, weight distribution, and structural integrity** for competition performance.
- **Authored technical documentation**, **delivered technical presentations**; coordinated with **multiple teams** and **external partners** for **project synergy**.

Publications

Paper Publication
International Journal of Science & Engineering Development Research
A Comprehensive Analysis of Two Innovative Aircraft Design Configurations

Bachelor's Thesis
JNTUH-CEH
CFD-Based Aerodynamic and Spray System Design for Enhanced Performance in Agricultural Drones

- **Spearheaded aerofoil design** with iterative **CFD analysis in ANSYS Fluent**, optimizing drone blade lift and controllability.
- **Engineered and simulated** a custom fertilizer spray nozzle using **CONVERGE**, performing internal/external flow simulations—methodology transferable to in-cylinder CFD for powertrain applications.

Experience

01/2023 – 03/2023
Hyderabad

Mechanical Design Intern
Garuda 3D

- **Contributed** to the **conceptualization** and **prototyping** of a **proof of concept** air taxi (scaled-down model) using **SolidWorks**, focusing on translating **innovative mobility solutions** into **tangible designs**.
- Executed **CFD simulations** on the **fuselage** using **ANSYS Fluent** and other tools, systematically studying **pressure distribution**, **drag**, and **lift** while iterating with multiple **aerofoil designs** to refine **performance**, evaluating and optimizing trade-offs between **stability**, **control**, and **efficiency**.
- Performed the **design, simulation, and fabrication** of **critical air taxi components** (e.g., wings, landing mechanism), specifically **optimizing** for **maneuverability** and **structural integrity** using a **Dual nozzle FDM based Rapid prototyping 3D printer (Raise 3D)**.

03/2023 – 07/2023
Hyderabad

Industrial Design Intern
Techolution

- **Led the end-to-end design and prototyping** of a **user-centric robot**, focusing on **robot dynamics** and **iterative design/simulation**.
- **Designed the robot's enclosure** ensuring **visual aesthetics**, **robust protection**, **optimal component integration**, and **ease of assembly**.
- **Designed and developed** a dedicated **test bench setup** to capture **visual data** for the purpose of training an **AI model** to guide a **robotic arm** in **screw-picking tasks**.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX
Pro Engineering

Simulation Software
Ansys Fluent
Converge
Openfoam (Basic)
Matlab - Simulink
Fusion 360
Solidworks

Software Proficiency
Python
C/C++
MS Office Suite

Data Visualization Tools
Power BI
Tableau

Languages

English
Fluent

German
Intermediate

S V V SAI SRI VALLABH PATANENI

Master's student in Automotive Engineering

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Summary

Innovative automotive engineer intent on shaping **tomorrow's urban mobility**. Fueled by a passion for **autonomy** and **smart systems**, I fuse deep hands-on experience in **mechanical design, simulation, prototyping, testing, and coding** skills sharpened through leading vehicle development teams and orchestrating cross-functional teams from concept to finished parts. At the nexus of **hardware and code**, I have crafted, simulated, and optimized complex **testbenches**, **sensor arrays**, and **control systems**, leveraging **Python** and **Simulink** to bring data-driven solutions to life. Now seeking a **mandatory internship at Robert Bosch GmbH, Renningen**, I am ready to turn real-world technical challenges into **robust, elegant answers** each project another step toward my vision of cities moved by **responsive, intelligent vehicles**.

Education

03/2026
Aachen, Germany

Master of Science in Automotive Engineering
RWTH Aachen University
Relevant Coursework: Automotive Engineering (Dynamics, Components and Assistant systems),Alternative Vehicle Propulsion systems, Mechatronics, Additive manufacturing.
GPA : 1.7

06/2019 – 04/2023
Hyderabad, India

Bachelors in technology - Mechanical Engineering
Jawaharlal Nehru Technological Univerisity
GPA: 1.3

Student Club

06/2024 – Present
Aachen, Germany

Ecogenium e.V
Mechanical Team lead & member of pit crew 24/25

- Currently **leading a multidisciplinary team** to **design, fabricate, and assemble** a **hydrogen-powered vehicle** from scratch for the **Shell Eco-Marathon**, managing the complete **lifecycle from concept to operational prototype**.
- **Suspension Test Bench**: **Designed and built** a dedicated setup to analyze **kinematics**, identify and improve upon **design weaknesses**, and validate **alignment** and **load distribution**.
- **Fuel Cell Test Bench**: **Engineered and programmed** a standalone rig for **operational testing (purging, short circuiting, telemetry)**; developed **automation routines** and **efficiency strategies** in **Python/Simulink**.
- **Led the complete design cycle**, **simulation**, **electronic integration**, and **manufacturing** of the vehicle's **suspension and steering system** using **cutting edge prototyping technologies**.
- Conducted **MIL simulations** in **Simulink** to optimize **control strategies** and validate vehicle performance, **leveraging competition data** to boost efficiency.
- **Authored technical documentation, delivered technical presentations**; coordinated with **multiple teams** and **external partners** for **project synergy**.

Academic Project

04/2025 – 07/2025

Automated Driving Practical - IKA

System integration Engineer

- **Developed sensor-driven testbenches** integrating radar and LiDAR to validate **Adaptive Cruise Control (ACC)** and **Lane Keep Assist (LKA) functions.**
- Developed models and **tuned PID controllers** using **MATLAB/Simulink and Python**, correlating simulation and real world results to optimize vehicle performance strategies through data-driven analysis, thereby advancing efficiency and control system effectiveness.
- Executed **Digital simulations** and **real time testing** using a mock vehicle and platforms like CARLA, directly linking lab validation with practical outcomes.
- **Comprehensive system integration:** Assembled, calibrated, and validated sensor arrays and control logic, bridging mechanical, electronic, and software domains for advanced driver assistance applications.

Experience

03/2023 – 07/2023
Hyderabad

Industrial Design Intern

Techolution

- Led the end-to-end **design and prototyping** of a user-centric robot, focusing on robot dynamics and iterative design/simulation.
- Developed and built a **custom test rig** to evaluate the **clamping performance** of a robotic **doorbot** across varied surfaces.
- **Tested** and compared **clamping strength** on **steel**, **aluminium**, and **carbon fibre** to optimize **material selection** and **reduce weight** while maintaining **reliability**.
- Integrated **test data** into the **prototyping process**, directly enhancing **mechanical design** and **operational safety**.
- **Designed and developed a dedicated test bench setup** to capture **visual data** for the purpose of training an **AI model** to guide a robotic arm in screw-picking tasks.

Skills

CAD Software
Fusion 360
SolidWorks
PTC Creo
Siemens NX

Software Proficiency
Python
C/C++
MS Office Suite

Simulation Software
Ansys
Matlab - Simulink
Fusion 360
Solidworks
Carla
ADAMS

Data Visualization Tools
Power BI
Tableau

Languages

English
Fluent

German
Intermediate

Recognitions and publications

Academic Recognition
RWTH
Acknowledged at the Semester Annual Meetup -RWTH for exemplary academic performance, achieving an outstanding overall grade of 1.2 in core automotive engineering courses.